Angle bisectors and perpendicular bisectors

ANSWER KEY

Core Concepts: Angle bisectors and perpendicular bisectors

1 In $\triangle ABC$ studying angle bisectors and perpendicular bisectors, $m\angle A = 41^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 94° ✓

b. 104°

c. 86°

d. 139°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (41^\circ + 45^\circ) = 94^\circ$.

2 A geometric figure scaled by factor $k = 2$ has an original side length of 5 cm. What is the corresponding image length?

a. 8 cm

b. 10 cm ✓

c. 15 cm

d. 20 cm

Multiplying the original length by the scale factor gives $5 \times 2 = 10$ cm.

3 On the coordinate plane, a line with slope $m = 2$ passes through $(0, 4)$. What is its y-coordinate when $x = 1$?

a. 4

b. 8

c. 6 ✓

d. 8

Substituting $x = 1$ into $y = 2x + 4$ gives $y = 2(1) + 4 = 6$.

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Procedural Fluency & Algebraic Manipulation

4 In $\triangle ABC$ studying angle bisectors and perpendicular bisectors, $m\angle A = 41^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 87° ✓

b. 97°

c. 79°

d. 139°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (41^\circ + 52^\circ) = 87^\circ$.

5 A geometric figure scaled by factor $k = 3$ has an original side length of 5 cm. What is the corresponding image length?

a. 12 cm

b. 15 cm ✓

c. 20 cm

d. 45 cm

Multiplying the original length by the scale factor gives $5 \times 3 = 15$ cm.

6 On the coordinate plane, a line with slope $m = 2$ passes through $(0, 4)$. What is its y-coordinate when $x = 2$?

a. 6

b. 10

c. 8 ✓

d. 10

Substituting $x = 2$ into $y = 2x + 4$ gives $y = 2(2) + 4 = 8$.

Graphical & Geometric Interpretation

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7 In $\triangle ABC$ studying angle bisectors and perpendicular bisectors, $m\angle A = 41^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 80° ✓

b. 90°

c. 72°

d. 139°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (41^\circ + 59^\circ) = 80^\circ$.

8 A geometric figure scaled by factor $k = 4$ has an original side length of 5 cm. What is the corresponding image length?

a. 16 cm

b. 20 cm ✓

c. 25 cm

d. 80 cm

Multiplying the original length by the scale factor gives $5 \times 4 = 20$ cm.

9 On the coordinate plane, a line with slope $m = 2$ passes through $(0, 4)$. What is its y-coordinate when $x = 3$?

a. 8

b. 12

c. 10 ✓

d. 12

Substituting $x = 3$ into $y = 2x + 4$ gives $y = 2(3) + 4 = 10$.

Exam-Style Problem Solving & Modeling

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10 In $\triangle ABC$ studying angle bisectors and perpendicular bisectors, $m\angle A = 41^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 73° ✓

b. 83°

c. 65°

d. 139°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (41^\circ + 66^\circ) = 73^\circ$.

11 A geometric figure scaled by factor $k = 5$ has an original side length of 5 cm. What is the corresponding image length?

a. 20 cm

b. 25 cm ✓

c. 30 cm

d. 125 cm

Multiplying the original length by the scale factor gives $5 \times 5 = 25$ cm.

12 On the coordinate plane, a line with slope $m = 2$ passes through $(0, 4)$. What is its y-coordinate when $x = 4$?

a. 10

b. 14

c. 12 ✓

d. 14

Substituting $x = 4$ into $y = 2x + 4$ gives $y = 2(4) + 4 = 12$.

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